

TrES-1b

R-1804

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_140904 · created 2026-08-03T02:38:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456904.7012 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.054
Transit depth (Rp/Rs)^2	0.00075 +/- 0.029 [%]
Orbital Inclination (inc)	86.59 +/- 1.46
Airmass coefficient 1 (a1)	0.161 +/- 0.093
Airmass coefficient 2 (a2)	1.61 +/- 0.66
Scatter in the residuals of the lightcurve fit is	55.19 %
Best Comparison Star	#1 - [265, 329]
Optimal Aperture	3.23
Optimal Annulus	6.37
Transit Duration (day)	0.07 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.003 $\pm$ 0.054 vs 0.1358 (deviation 2.46 σ)
Duration fit vs published	0.07 d vs 0.1042 d (deviation 1.42 σ)
Mid-time O-C	-2.7 min
Depth SNR	0.1
Residual scatter	55.19 %

Light curve

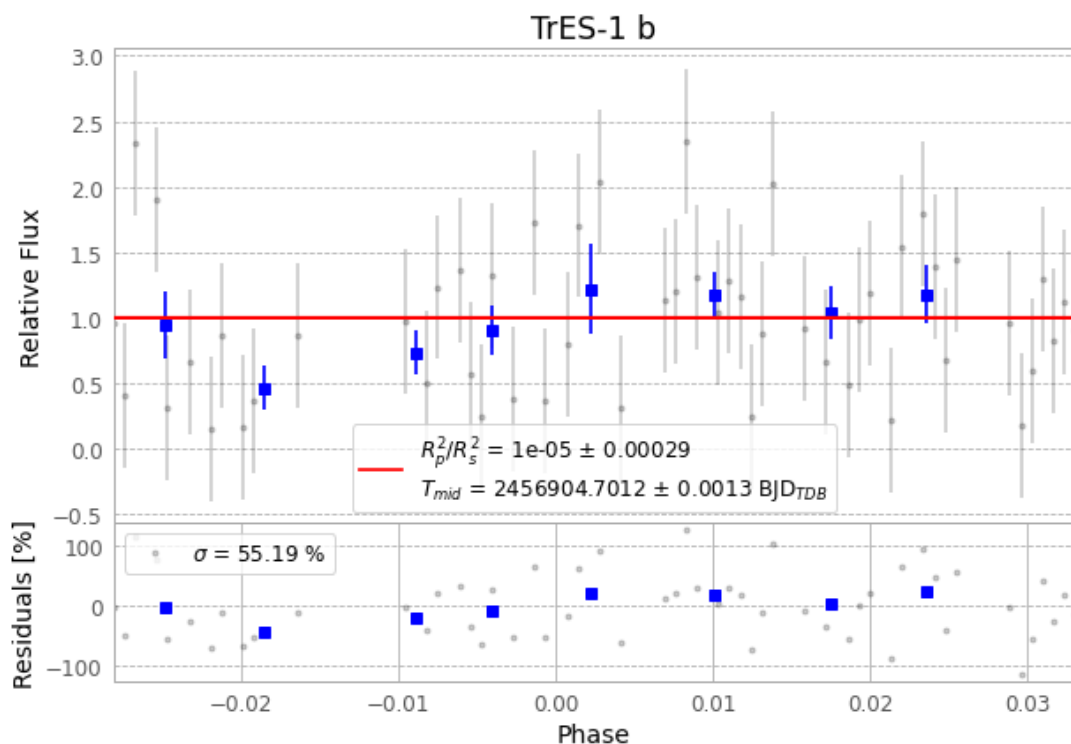


Figure 1 — FinalLightCurve_TrES-1 b_2014-09-03.png (SHA-256 69ccef7d340a5f8c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [604, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 848aad949e5678b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), TRES-1140904024212.FITS → TRES-1140904070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140904.log (54094ca8264a...), FinalLightCurve_TrES-1 b_2014-09-03.png (69ccef7d340a...), FinalLightCurve_TrES-1 b_2014-09-03.csv (f4cd8069647c...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 02:38Z.